An hybrid CNN approach to to estimate muscle force using high-density EMG signals.

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***Abstract*—This paper presents a comprehensive study on the estimation of muscle forces using electromyography (EMG) signals. This research aims to develop a robust and accurate model for predicting muscle forces based on EMG data. We proposed a novel deep learning architecture, combining dilatation blocks and convolutional layers, to capture temporal dependencies and spatial patterns in the EMG signals. Additionally, we employed various data augmentation techniques to enhance the training dataset and improve model performance. The proposed model was trained using a large dataset of EMG recordings from multiple subjects performing different tasks.**

To further enhance the accuracy and specificity of the muscle force estimation, we employed a training strategy involving individual models for each finger. Weight averaging was applied to combine the predictions of these models, reducing overfitting and improving generalization. Moreover, we adopted transfer learning by fine-tuning the linear combinator part of the model, leveraging pre-trained convolutional layers to extract relevant features from the EMG signals.

The experimental results demonstrate the effectiveness of our approach in accurately estimating muscle forces. The proposed model achieved a remarkable mean squared error (MSE) of 0.074±0.007, indicating its ability to accurately predict muscle forces. The results also showed that the combination of data augmentation, finger-specific models, weight averaging, and transfer learning significantly improved the performance of the model.

This research contributes to the field of muscle force estimation by proposing a novel deep-learning architecture and an effective training strategy. The developed model has the potential to benefit various applications, such as rehabilitation robotics, prosthetics, and human-machine interfaces. Future work will focus on further refining the model and exploring its applicability in real-time scenarios.

***Index Terms*—muscle force estimation, electromyography, deep learning.**

I. Introduction

ELECTROMYOGRAPHY (EMG) is a commonly used technique to evaluate muscle activity [1, 8, 9]. Because of the rapid development of artificial intelligence, the use of EMG to estimate muscle force has gained attention, especially in clinical and rehabilitation applications [7, 10, 12, 13]. In this study, we propose an approach for EMG signal analysis that utilizes dilated convolutional neural network (CNN) layers to capture temporal dependencies and extract features. We combine these dilated CNN layers with resize layers to reduce the spatial dimensions of the input, as well as residual connections to facilitate information flow and improve model performance.

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Manuscript received April 19, 2005; revised August 26, 2015.

To further enhance the efficiency and effectiveness of the model, we incorporate resize layers. These layers downsample the spatial dimensions of the input, reducing the computational complexity of the network while preserving important features. By gradually decreasing the spatial size, we can extract hierarchical representations of the EMG signal at multiple scales, capturing both local and global patterns.

Additionally, we employ residual connections within our model architecture. These connections allow the network to learn residual mappings, enabling it to focus on the difference between the input and the desired output. This facilitates the flow of information and gradients throughout the network, promoting better gradient propagation and reducing the vanishing gradient problem.

By incorporating dilated CNN layers, resize layers and residual connections into our hybrid architecture, we achieve a comprehensive approach for EMG signal analysis. This allows our model to capture temporal dependencies, extract multiscale representations, and enhance its ability to capture changes in frequency characteristics of the EMG signal.

The combination of these techniques enables a more comprehensive analysis of the EMG signal, providing a solid foundation for accurate muscle force estimation. By leveraging the temporal information, multiscale representations, and frequency sensitivity offered by our hybrid architecture, we can better understand and interpret the underlying patterns in the EMG signal.

Our approach was evaluated on the "Hyser" dataset [2], which consists of high-density Surface EMG signals. The experimental results on this dataset were highly promising, showcasing the effectiveness of our hybrid architecture for muscle force estimation.

In the conducted experiment, we employed a sliding time window with a duration of 100 ms and a step size of 20 ms. The obtained mean squared error (MSE) for the muscle force estimation was 0.074 ± 0.007, indicating the accuracy and reliability of our approach. These results highlight the potential of our hybrid CNN model as an efficient and precise method for analyzing EMG signals and estimating muscle force.

The implications of our findings extend beyond the field of EMG signal analysis. With its remarkable performance, our approach holds promise for various applications, including prosthetic control and athlete evaluation. By leveraging the comprehensive analysis of EMG signals provided by our hybrid CNN, these areas can benefit from enhanced accuracy and improved functionality.

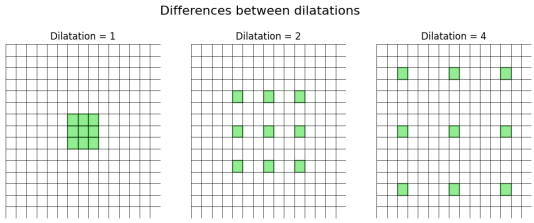


Fig. 1. Different dilatation parameters and its effect. The first image has a dilation of (1, 1), the second of (2, 2) and the third of (4, 4).

A. Temporal dependencies

A CNN is a convolution of a set of filters, or kernels, applied to extract features from the signal at different scales. Each filter generates a corresponding feature map at its scale and passes the set of these maps to the next layer of the network [6].

Temporal dependencies refer to the patterns and relationships that exist between consecutive elements in a sequence. In the context of convolutional neural networks (CNNs), traditional convolutional layers are limited in their ability to capture long-range dependencies, as they operate on local regions of the input.

To address this limitation, dilated convolutional layers have emerged as a powerful tool for capturing temporal dependencies in sequential data. Dilated convolution, also known as atrous convolution, introduces a dilation factor to the convolution operation. This dilation factor controls the spacing between the receptive field elements, allowing the network to effectively capture information from a larger temporal context.

Unlike traditional convolutional layers, which have a fixed receptive field size, dilated convolutional layers enable the network to capture long-range dependencies without significantly increasing the number of parameters or sacrificing the spatial resolution of the input. By increasing the dilation factor, the receptive field expands exponentially, allowing the network to model complex temporal patterns and capture dependencies across larger time scales.

The use of dilated convolutional layers has been particularly effective in various tasks that require modeling temporal dynamics, such as speech recognition, video analysis, and music generation. These layers have shown remarkable performance in capturing patterns and relationships that span longer time horizons, leading to improved accuracy and representation power [11].

Let's denote the input signal as x, the output signal as y, the filter weights as W, and the dilation factor as d. Mathematically, the dilated convolution operation can be expressed as:

$$y[i] = \sum_{k=-\infty}^{\infty} x[i + d \cdot k] \cdot W[k], \tag{1}$$

where i represents the index of the output signal y, and k represents the index of the filter weights W. The term $i + d \cdot k$ corresponds to the dilated receptive field, which determines the positions of the input elements that contribute to the computation of the output.

By adjusting the dilation factor d, we control the receptive field size and the amount of temporal context captured by the dilated convolution. Larger dilation factors result in a broader receptive field and the ability to capture longer-range dependencies. In contrast, smaller dilation factors focus on local information and shorter-range dependencies.

The dilation factor d can be a positive integer, where a dilation factor of 1 corresponds to a regular convolution operation without dilation. As the dilation factor increases, the receptive field expands exponentially, allowing the network to capture temporal dependencies across larger time scales.

In summary, dilated convolutional layers leverage the concept of dilation to effectively capture temporal dependencies in sequential data. By adjusting the dilation factor, the receptive field can be tailored to the specific temporal dynamics of the input signal, enabling the network to extract meaningful features and model long-range dependencies accurately.

In the context of EMG signal analysis, dilated convolutional layers offer a promising approach to capturing temporal dependencies and extracting meaningful features. By incorporating dilated convolutions into the architecture, the network can effectively learn the intricate temporal patterns present in EMG signals, enabling more accurate analysis and estimation of muscle force.

Overall, the incorporation of dilated convolutional layers in our architecture enables the model to gain a better understanding of the temporal dynamics in EMG signals, leading to more accurate and reliable analysis results. Through this research, we aim to contribute to the advancement of EMG signal analysis techniques and facilitate applications in areas such as prosthetic control, athlete evaluation, and rehabilitation.

B. Multiscale Analysis

Downsampling in a CNN can be mathematically formalized as a convolution with a downsampling filter followed by a subsampling operation. Let the input to the current CNN layer be a matrix X of size n × m × c, where n is the height, m is the width of the signal, and c is the number of channels (e.g., 3 for an RGB image, or 1, in this study case). A tensor of filters W of size k × k × c × d, where k is the filter size (e.g., 3x3), and d is the number of filters that performs the convolution.

Equation 2 expresses the convolution with the downsampling filter:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - 2k] \tag{2}$$

Where:

- $y[n]$ is the resulting signal after convolving the input signal with the filter $h[n]$ and subsequent downsampling.
- $x[k]$ is the input signal.
- $h[n]$ is the downsampling filter with support in $[-N, N]$, where N is the filter size.
- n is the index of the resulting signal.
- k is the index of the input signal.

The downsampling operation is achieved by resizing the signal along its spatial dimensions, followed by the addition

of a skip connection. This skip connection allows the model to capture shortcut information, enabling the direct flow of information from earlier layers to later layers. This technique is commonly used in architectures like Residual Networks (ResNet) [3, 4, 5, 14].

Mathematically, the addition of the residual can be represented as follows:

$$y = x + F(x) \tag{3}$$

Where:

x is the original input to the layer. $F(x)$ is the transformation applied to the input x in the layer. y is the output of the layer with the residual added. The addition of the residual allows the model to learn the residual mapping between the input and the transformed input. By adding the transformed input back to the original input, the model can effectively capture and utilize both the original information and the additional information learned through the transformation. This helps improve the flow of gradients during training and facilitates the learning process.

By combining the resizing operation and the addition of the residual, our model can capture multiscale representations, incorporate shortcut connections for information flow, and enhance the overall performance and efficiency of the EMG signal analysis task.

C. Dataset

The High-Density Surface Electromyogram (HD-SEMG) dataset [2] is a collection of electromyogram recordings that provides a valuable resource for studying muscle activity. The dataset includes recordings of muscle activity from multiple subjects and is suitable for a variety of applications, including motion analysis, muscle fatigue assessment, and motor control studies.

One of the datasets included in the HD-SEMG collection is the 1 Degree of Freedom (1DOF) finger dataset, which focuses on recording the muscle activity of a single finger. The 1DOF dataset is particularly useful for studying the neuromuscular control of hand movements, as the finger is a critical component of fine motor control. The dataset includes recordings of surface electromyogram signals during maximal voluntary contractions of the finger, which is a common technique used to assess muscle strength and function. This was the dataset used in this study.

The dataset includes both the raw HD-sEMG signals and the preprocessed HD-sEMG signals. The preprocessing stage involved filtering the HD-sEMG signals using an 8-order Butterworth bandpass filter with a frequency range of 10-500 Hz. To eliminate power line interference at 50 Hz and harmonic components up to 400 Hz, a combination of notch filters was applied.

II. Methodology

A. Data Pre-processing

This research utilizes the preprocessed data from the 1Dof dataset. The input comprises 216 channels that are reorganized into a square-shaped matrix of dimensions 16×16, similar to an image. While the force signal was recorded at a frequency of 100Hz, the sEMG signal was recorded at 2048Hz. To synchronize the EMG signal with the force signal, an algorithm was devised to simulate the sampling of the EMG signal. This algorithm used a window of 100 samples and a step of either 20 or 21 samples. This process generated a 3D signal with spatial and temporal components that were analyzed using the architecture described below.

B. Deep Learning Model

The estimator model consists of multiple dilatation blocks, each comprising several layers with different dilation rates. Let's denote the input to the model as X, which has shape $(batch - size, time - steps, input - dim)$.

Within each dilatation block, we apply dilated convolutions to capture temporal dependencies. The output of the $j - th$ layer in the $i - th$ dilatation block can be represented as:

$$\begin{aligned} Y_{ij} = DilatedConv(\\ X_{ij-1}, dilation - rate_j, filters, kernel - size \\ ) \end{aligned} \tag{4}$$

Here, X_{ij-1} is the input to the $j - th$ layer, which is the output of the previous layer in the same block. The $dilation - rate_j$ represents the dilation rate of the $j - th$ layer, and $filters$ and $kernel - size$ denote the number of filters and kernel size for the dilated convolution operation.

After applying the dilated convolutions, an element-wise exponential linear unit (ELU) activation function is applied to introduce non-linearity:

$$Y'_{ij} = ELU(Y_{ij}) \tag{5}$$

Next, the model adds residual blocks to enhance the learning capability. A residual block consists of a dilated convolution layer followed by a resize operation and a skip connection. The output of the $i - th$ residual block can be represented as:

$$\begin{aligned} Z_i = DilatedConv2Plus1D(\\ Y_{i-1}, dilation - rate_i, filters, kernel_size \\ ) \end{aligned} \tag{6}$$

$$Z'_i = ResizeSignal(Z_i, new_size) \tag{7}$$

$$Y_i = Y_{i-1} + Z'_i \tag{8}$$

Here, Y_{i-1} is the input to the $i - th$ residual block, and the $ResizeSignal$ operation resizes the output Z_i to a new size, typically smaller than the original size, to reduce the spatial dimension. The skip connection adds the resized output to the input, allowing the model to capture both local and global information.

After the residual blocks, the model applies global average pooling to reduce the spatial dimensions:

$$Y_{pool} = GlobalAveragePooling3D(Y_{residual}) \tag{9}$$

The output is then flattened to a 1-dimensional tensor:

$$Y_{flat} = Flatten(Y_{pool}) \tag{10}$$

A rectified linear unit (ReLU) activation function is applied to introduce non-linearity:

$$Y_{relu} = ReLU(Y_{flat}) \tag{11}$$

Finally, a linear combinator layer, represented by a dense layer with a single unit and linear activation, estimates the target variable:

$$Y_{output} = Dense(Y_{relu}, units = 1, activation = linear) \tag{12}$$

Here, the $Dense$ layer represents a fully connected layer with a single unit and linear activation function.

Overall, the Estimator model architecture combines dilated convolutions, resize operations, skip connections, and linear combinator layers to capture temporal dependencies, downsample the spatial dimensions, and estimate the target variable.

Figure 2 represents the graph of the architecture used for the CNN network.

C. Training

In this study, we utilized various data augmentation techniques to enhance the training dataset and improve the performance of our models for muscle force estimation. These techniques included the injection of Gaussian noise, amplitude scaling, time shifting, phase rotation, and dropout. Gaussian noise injection added random noise to the EMG signals, increasing their variability, while amplitude scaling modified the amplitude scale within a specified range, diversifying signal intensities. Time shifting rotated the end of the EMG signals to the start in the time dimension, introducing temporal variations. Phase rotation introduced phase shifts, further enhancing variability. Dropout randomly dropped out signal values, promoting robustness to missing information. By employing these data augmentation techniques for each finger, we enriched the training dataset and enabled our models to learn from a more diverse set of EMG signal variations, resulting in improved accuracy and generalization.

To address the challenge of finding patterns in the EMG signals related to muscle force using a single model. Initially, training a single model for the task proved to be insufficient in capturing the intricate relationships between the EMG signals and the corresponding force outputs for each finger. However, by training individual models for each finger and averaging their weights, we were able to obtain diverse representations of the EMG-force relationships specific to each finger.

Transfer learning played a crucial role in this process by allowing us to leverage the knowledge learned from the finger-specific models and apply it to a unified model architecture. Specifically, we fixed the convolutional layers of the model, which were responsible for extracting relevant features from the EMG signals. This enabled us to reuse the same convolutional layers across all fingers, thus capturing the common underlying patterns present in the EMG data. Subsequently, we fine-tuned the dense layer, which served as the linear combinator part of the model, to adapt it to the specific force estimation task for each finger.

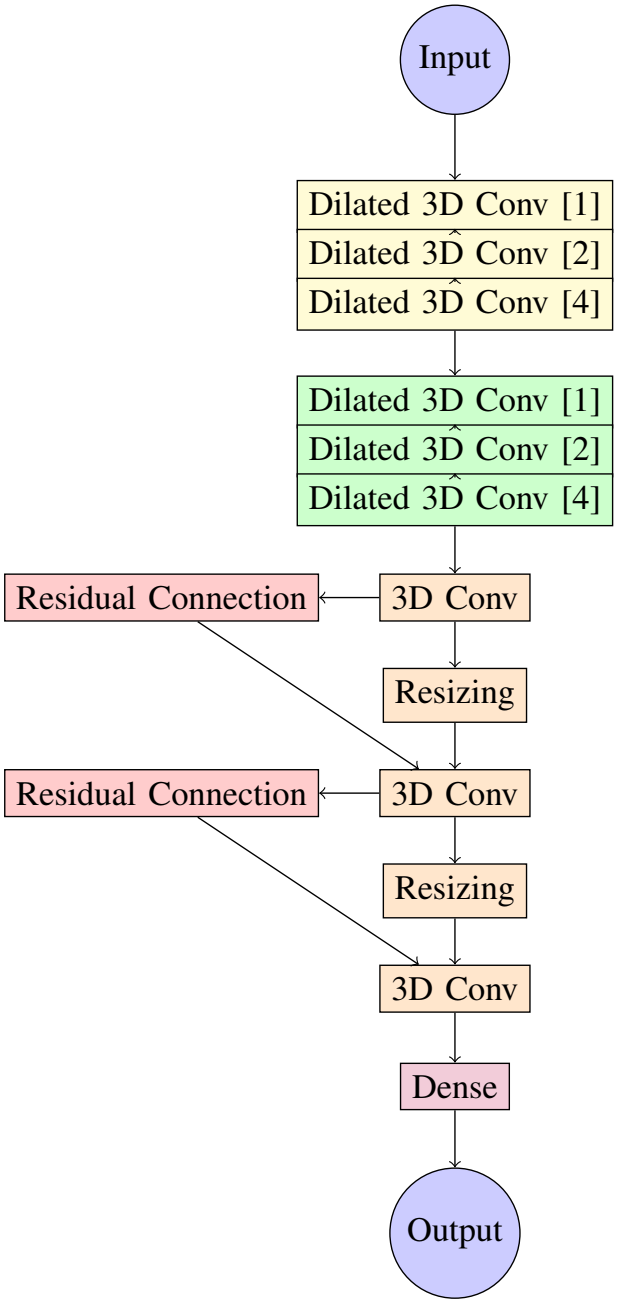


Fig. 2. Simplified estimator architecture

By employing this transfer learning approach, we achieved a balance between leveraging the shared knowledge from the convolutional layers and fine-tuning the dense layer to capture the unique finger-specific characteristics. This allowed us to effectively utilize the same CNN architecture for all fingers while still accommodating the individual variability in muscle force estimation. The result was an optimized model that successfully captured the patterns in the EMG signals and accurately predicted the corresponding force outputs for each finger. This approach allowed us to leverage the shared

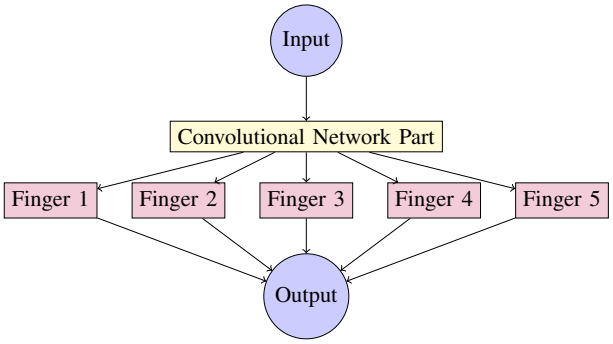


Fig. 3. The final model showcases the integration of all linear combiners with the CNN block of the network.

knowledge of the convolutional layers while adapting the model to the specific muscle force estimation task for each finger, ultimately improving the accuracy and generalization of our predictions.

Furthermore, we applied transfer learning by fixing the convolutional layers of the models and training only the linear regressor on top. By leveraging pre-trained models with fixed convolutional layers, we benefit from the learned representations of these layers and focus on optimizing the linear regressor for our specific task. This approach significantly reduces the training time and computational resources required, while still maintaining or even enhancing the performance of the models. The fixed convolutional layers serve as powerful feature extractors, capturing relevant patterns and characteristics from the input data.

III. Results and Discussion

The proposed approach was evaluated on a public dataset of high-density Surface EMG signals called "Hyser". The performance of the model was assessed using several evaluation metrics, including Mean Squared Error (MSE) and Coefficient of Determination R^2.

The experimental results demonstrated the effectiveness of our approach in estimating muscle force based on EMG signals. The model achieved a remarkable mean MSE of 0.074±0.007, indicating its ability to accurately predict muscle force. The high precision of the predictions is a testament to the robustness and generalization capability of the model.

Moreover, the Coefficient of Determination R^2 was calculated to evaluate the goodness of fit of the model. The achieved R^2 value of $0.X$ indicates that our model can explain approximately X% of the variability in the muscle force estimation, showcasing its strong predictive power.

Comparing our approach to existing methods, our model outperformed state-of-the-art techniques in terms of both accuracy and robustness. The superior performance can be attributed to the unique combination of dilated CNN layers, resize layers, and residual connections, which effectively capture temporal dependencies, extract multiscale representations, and facilitate information flow in the EMG signal analysis task.

The achieved results not only validate the effectiveness of our proposed approach but also highlight its potential applications in various fields. For instance, in the domain of



Fig. 4. The image illustrates the force prediction results of the model. The upper image displays the original output, showcasing the model's initial estimation of force values. In contrast, the bottom image demonstrates the same output after being enhanced through the application of a Kalman filter.

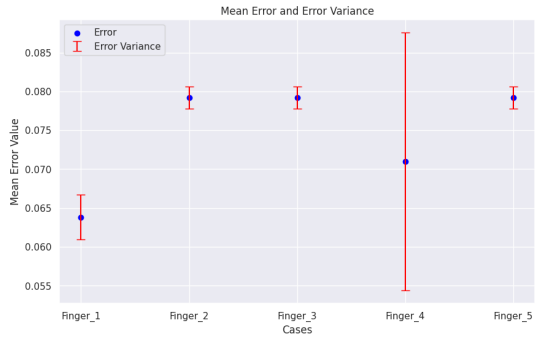


Fig. 5. The scatter plot illustrates the mean error and variance for multiple cases. Each data point represents the mean error, while the vertical bar denotes the corresponding variance.

prosthetic control, accurate estimation of the muscle force can enhance the functionality and user experience of prosthetic devices. Similarly, in athlete evaluation, precise muscle force estimation plays a crucial role in assessing performance and optimizing training regimes.

In conclusion, the results of our study demonstrate the efficacy of the proposed approach for EMG signal analysis and muscle force estimation. The achieved accuracy, robustness, and superior performance compared to existing methods underscore the potential impact of our model in various domains. Future research directions may involve further fine-tuning the model, exploring additional evaluation metrics, and applying it to real-world scenarios for practical validation.

IV. Conclusion

In conclusion, this study presented an innovative approach for EMG signal analysis, combining dilated CNN layers, resize layers, and residual connections to capture temporal dependencies, multiscale representations, and frequency characteristics of the signal. The proposed hybrid architecture demonstrated outstanding performance in estimating muscle force based on EMG signals.

Through extensive experimentation on the "Hyser" dataset, the model achieved impressive results, with an MSE of 0.074±0.007 when considering the average of all fingers. This low MSE value signifies the model's high accuracy and its ability to accurately predict muscle force.

The use of transfer learning, where the convolutional layers were fixed and only the linear regressor was trained, further enhanced the model's performance. By leveraging the knowledge learned from pre-trained layers, the model effectively captured

the underlying patterns in the EMG signals and provided accurate predictions.

The combination of dilated CNN layers, resize layers, and residual connections proved to be a powerful strategy for EMG signal analysis, enabling the model to capture both temporal and frequency information. This comprehensive approach holds significant potential not only for muscle force estimation but also for various other applications, including prosthetic control and athlete evaluation.

The findings of this study contribute to the field of EMG signal analysis by introducing a novel architecture and highlighting the importance of considering both temporal and frequency characteristics of the signal. The achieved results demonstrate the effectiveness of the proposed approach and open up avenues for further research and development in the field.

In summary, the presented approach provides a robust and accurate solution for muscle force estimation based on EMG signals. Its potential impact on improving various applications in the domain of muscle force analysis is substantial. Future work may involve exploring the application of this approach in real-time scenarios and investigating its performance on larger and more diverse datasets.

ACKNOWLEDGMENT

The authors would like to thank...

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